What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Antiretrovirals: Drugs to Treat and Prevent HIV Infection

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Adapted from: Drugs to Treat and Prevent HIV Infection

Listen to this Brick

0:00 / 24:13 1x

Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Describe the mechanism of action and mechanism of resistance of antiretroviral agents used in the treatment and prevention of HIV infections.
2. List the indications for antiretroviral agents used in the treatment and preventi
3. Recognize the common adverse reactions associated with drugs used in the treatment and prevention of HIV infections.
4. Identify the common drug interactions associated with drugs used in the treatment and prevention of HIV infections.

CASE CONNECTION

MB, a 40-year-old nurse, has sustained a needlestick injury while on duty. The hollow-bore needle was contaminated with blood, and there is specific reason to be concerned that the source patient is HIV positive. MB says, "He is in the midst of getting diagnosed. They just sent his blood tests a couple of hours ago, separate from this. Clinically, it looks like he has AIDS. I'm scared." MB is generally healthy and not taking any medications. The puncture wound has already been scrubbed.

What will you say to MB about whether he should take anti-HIV medications at this time? Consider your answers as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

What Are HIV Therapeutics and Prophylaxis?

The retrovirus human immunodeficiency virus (HIV), while incurable, has become quite manageable. Nevertheless, an ounce of prevention is worth a pound of cure, as they say, and this is especially true of HIV.

Antiretroviral drugs c of therapy and even prophylaxis. So what are these drugs and what do we need to know about them? The antiretroviral drugs act on specific steps of the retroviral life cycle. In this brick we will focus on learning about the five different drug classes used to manage HIV, **later, in the organ system courses, you will learn specific medications.**

How do Antiretroviral Drugs work? (LO#1)

The mechanism of action for specific antiretroviral drugs varies depending on the class and which viral site the drug acts on (Figure 1). In addition, some drugs are administered with pharmacokinetic boosters, or enhancers, to improve their effectiveness. Now, let's dive into the mechanisms for each type of antiretroviral drug. Make sure to refer back to the figure throughout our discussion.

HIV_Condensed

Entry/Fusion Inhibitors

For HIV to enter a host cell, it must bind to a receptor on the surface and then fuse. One of these receptors is CCR5 (Figure 2). Maraviroc blocks CCR5, which prevents binding of the virus, causing it to be unable to enter the cell. This drug is sometimes useful for patients with drug-resistant virus infection. Enfuvirtide binds to gp41, which becomes exposed once HIV binds gp120, blocking fusion of the virus and its entry into the cell.

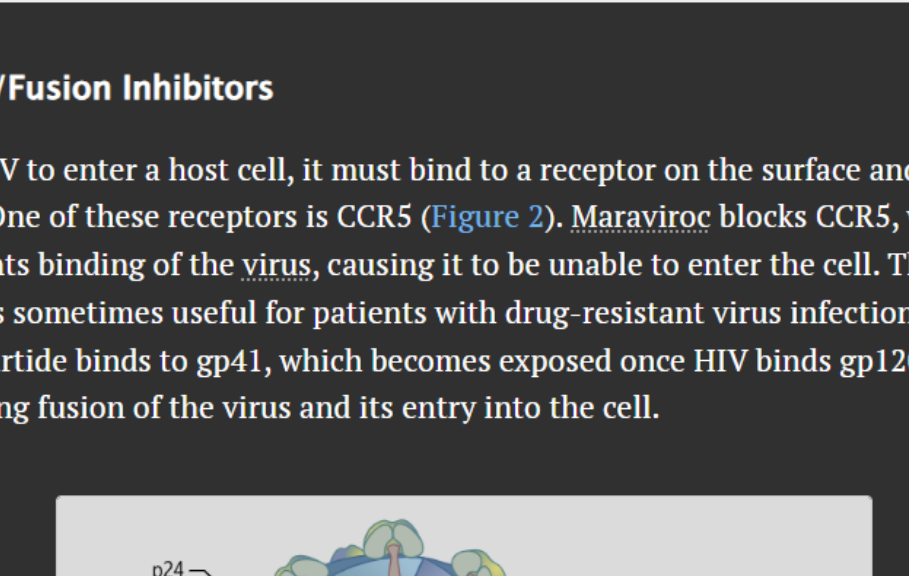

Figure 2

Nucleotide/Nucleoside Reverse Transcriptase Inhibitors

To replicate, retroviruses use the enzyme reverse transcriptase to convert RNA to DNA. Without this step, a virus such as HIV cannot reproduce. Nucleotide/nucleoside reverse transcriptase inhibitors (NRTIs) inhibit this enzyme. They may be derived from adenosine (tenofovir), cytosine (emtricitabine, lamivudine), guanosine (abacavir), or thymidine (zidovudine, or AZT). They are chain terminators, which are incorporated into the growing nucleotide chain, competing with the individual nucleotides. This stops reverse transcriptase. These drugs, therefore, are a mainstay of treatment.

Non-Nucleoside Reverse Transcriptase Inhibitors

Non-nucleoside reverse transcriptase inhibitors (NNRTIs), like NRTIs, inhibit reverse transcriptase, rendering the elongating DNA strand incomplete and useless. They bind directly to reverse transcriptase, inactivating it. This stops the DNA chain from polymerizing. NNRTIs are often used with NRTIs, especially tenofovir.

Integrase Strand Transfer Inhibitors

Normally, newly made viral DNA must be integrated into the host cell's genome. This is catalyzed by the enzyme integrase. Integrase strand transfer inhibitors (INSTIs) are drugs that inhibit integrase, preventing this DNA from infecting cells, which very common in most HIV medication regimens.

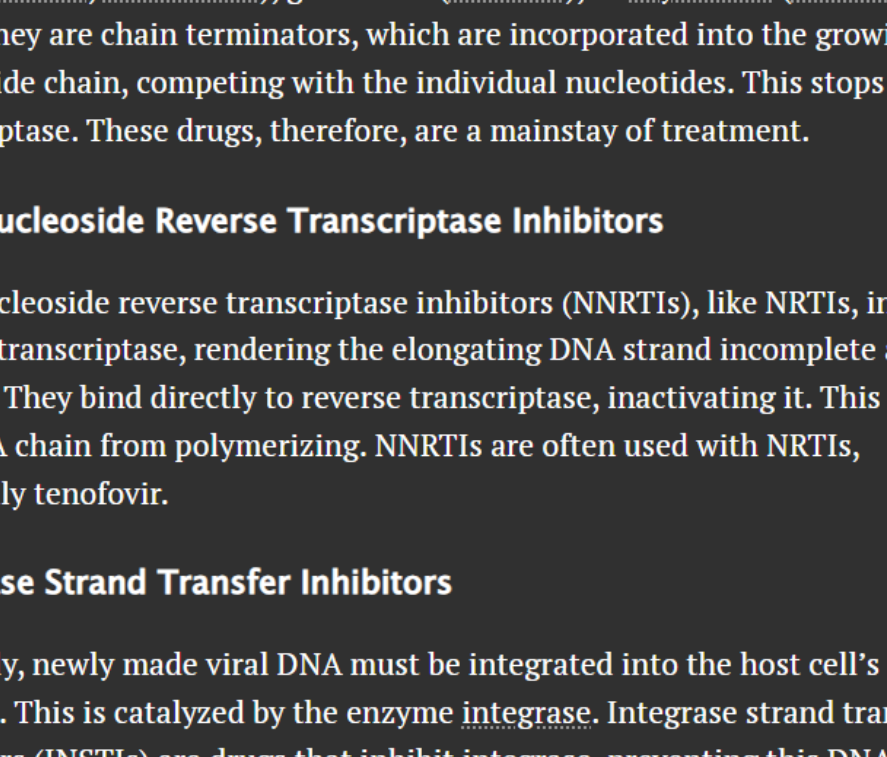

Figure 3

Protease Inhibitors

These drugs inhibit the final step of replication, which involves budding of the virus from the host cell. Protease is an enzyme that is responsible for cutting polyprotein chains into smaller proteins that are mature HIV units. By inhibiting this enzyme, protease inhibitors stop mature viral particles from forming. They are often considered when a patient is not responding to initial antiretroviral therapy

Pharmacokinetic Enhancers

Sometimes, additional drugs are used alongside antiretrovirals to increase the potency or efficacy of the drug. Specifically, protease inhibitors often require a booster with *ritonavir* (a protease inhibitor itself) or *cobicistat. These boosters inhibit hepatic cytochrome P450 (CYP450), increasing the concentration of the protease inhibitor.* This has the benefit of increasing patient adherence by decreasing the frequency with which patients must take their medication. It also limits the adverse effects seen at higher doses. In addition, the integrase inhibitor elvitegravir may require boosting.

CLINICAL CORRELATION

Do all HIV drugs require a pharmacokinetic booster? As these enhancers, ritonavir or cobicistat, inhibit CYP enzymes only a handful of medications require this boost, and it is meant to prolong the duration of the drug to decrease the dosing frequency (i.e., the patient doesn't have to take it as often). This helps with adherence to the medication!

The most common HIV drugs that need boosters are the protease inhibitors Atazanavir, Darunavir, and Lopinavir, and the INSTI Elvitegravir.

Figure 4 includes many of the HIV medications/ classes that are in use today.

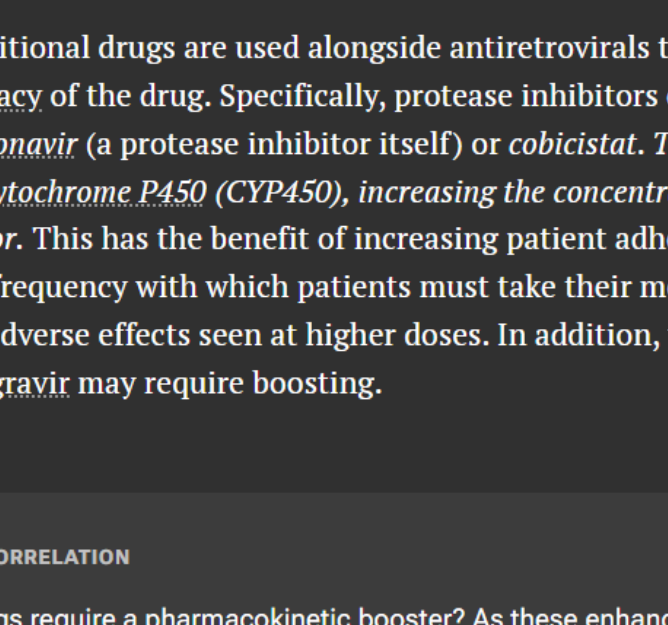

Figure 4

Resistance

Recall from the "Retroviridae Part 2 of 2" brick, that HIV is difficult to cure due to two key steps in its replication pathway: the lack of proofreading by reverse transcriptase, leading to rapid mutation and adaptation, and the incorporation of viral DNA into the host genome. The high mutation rate of HIV allows it to quickly adapt to treatments, necessitating the use of multiple drugs simultaneously to increase the number of mutations needed for resistance.

What is the mechanism of action of protease inhibitors?

How Are Antiretroviral Drugs Used in the Clinical Setting? (LO#2)

In a patient infected with HIV, a combination of antiretroviral drugs is preferred. A multidrug treatment plan has been clinically shown to best suppress HIV RNA. *This combination method is the mainstay of highly active antiretroviral treatment (HAART) and can slow or halt the progression to acquired immunodeficiency syndrome (AIDS).*

Which classes of drugs, then, would we select to initiate therapy? Many studies have supported the initial use of two NRTIs plus one drug from another class, most commonly an INSTI. In some cases, the third drug may be a protease inhibitor or an NNRTI. This type of multidrug regimen can reduce morbidity and mortality related to the infection. In the case of a drug-resistant virus, treatment may need to be more selective. For example, NNRTIs are the most common class in the United States to be ineffective because of resistance. In this case, another type of antiretroviral should be considered in addition to NRTIs.

CLINICAL CORRELATION

We have talked about the drugs/ classes individually, so far, but, as mentioned above these medications are ALWAYS used in combination. This helps reduce resistance rates. To ease the burden of taking 3-4 tablets at a time, and likely increase regimen adherence, many combination tablets are available to patients. When prescribing these medications it is best, in general, to have the patient take as few medications as possible! There are so many combos (Don't worry about memorizing them), but a few of the most common are listed below:

Bictegravir, emtricitabine, and tenofovir

Dolutegravir, abacavir, and lamivudine

Emtricitabine and tenofovir

Dolutegravir and lamivudine

Which Drugs Are Used in HIV Prophylaxis?

In addition to using antiretrovirals for therapy, pre-exposure and postexposure prophylaxis are good ways to implement antiretrovirals for infection prevention.

Postexposure Prophylaxis

In the case of accidental exposure, antiretroviral therapy is also used. This is called postexposure p en has shown the most promise, especia be discontinued if testing shows the patient to be noninfected. For non–health care workers, nonoccupational postexposure prophylaxis (nPEP) is used. This also includes antiretroviral therapy within 72 hours of exposure. It should be considered for people who have had unprotected sex or percutaneous blood exposure.

Pre Exposure Prophylaxis

Certain people, such as intravenous (IV) drug users, sex workers, and men who have sex with men (MSM), are at higher risk for HIV infection. For these at-risk populations, PrEP NNRTI/NRTIS combo can be offered. Antiviral therapy, in addition to education and encouraging condom use, should be used after determining risk based on sexual and substance use history. A detailed history is important to determine risk, and raising awareness in high-risk groups as well as clinicians can increase the amount of people willing to take medication. This awareness may be increased by focusing on clinics and centers that receive a large number of young people and IV drug users.

Opportunistic Infection Prophylaxis

Other than antiretroviral treatment, how can the opportunistic infections of AIDS be prevented? Commonly, trimethoprim-sulfamethoxazole (TMP-SMX) is used to prevent *P jiroveci* pneumonia for a CD4 count <200/mm^3 and toxoplasmosis for counts <100/mm^3. Once the CD4 count goes lower than 50/mm^3, *Mycobacterium avium* complex may be prevented with azithromycin, although new guidelines suggest that antiretroviral therapy alone is enough.

CLINICAL CORRELATION

AIDS is generally diagnosed with the presence of AIDS-defining illnesses (such as *Pneumocystis jiroveci* pneumonia) and CD4 cell count <200/mm^3. Using HAART can keep CD4 cell count up by suppressing the viral RNA and reducing the inflammatory immune response.

What is PrEP used for?

HIV Therapy in Pregnancy

Does therapy protocol change in the pregnant patient? It does, in the following way: for the NRTI component, the drugs of choice are tenofovir disoproxil fumarate (tenofovir DF) and emtricitabine or abacavir and lamivudine. The third drug is generally a protease inhibitor (first trimester) or integrase inhibitor (after the fi n during pregnancy can reduce the mother-to-child transmission to less than 2%. Safety for the fetus must also be taken into account. For pregnant patients at risk for preterm birth, protease inhibitors are generally avoided. In addition, dolutegravir may be continued in pregnant patients already taking it, but it is sometimes avoided in newly diagnosed patients because of the risk for neural tube defects.

What are the adverse effects of antiretrovirals? (LO#3)

In this brick we have discussed drug classes of antiretrovirals. As HIV is covered in more detail later in the organ systems, specific medications within each class will be discussed. For the adverse effect profiles we will call out important medications/ adverse effects from each class.

Nucleotide/Nucleoside Reverse Transcriptase Inhibitors

Important adverse effects to remember when dealing with NRTIs have to do with mitochondrial toxicity, because of the drugs' inhibition of human DNA polymerases as well as HIV's reverse transcriptase. This can present as pancreatitis or neuropathy. Tenofovir may cause kidney injury or bone loss, zidovudine may cause bone marrow suppression (neutropenia), and didanosine may cause pancreatitis. In fact, didanosine is rarely used because of the risk of complications.

Recall from our discussions on pharmacogenomics that abacavir has a unique hypersensitivity reaction. All patients who are HIV positive MUST HAVE genetic testing to determine if the HLA-B*5701 allele is present, regardless of whether that will be a part of their medication regimen or not.

CLINICAL CORRELATION

Tenofovir is one of the most common HIV medications, but there are two forms of this medication, Tenofovir disoproxil fumarate (TDF) and tenofovir alafenamide (TAF). Mechanistically, they work the same. Both are prodrugs, however, TDF results in much higher plasma rates and is much more likely to cause the renal injury and bone loss mentioned above. The newer version, TAF, has a significantly lower rate of these adverse effects and is the preferred version to use. Nowadays, when referring to tenofovir, it is implied that you are talking about TAF.

Non–Nucleoside Reverse Transcriptase Inhibitors

The NNRTI efavirenz must be monitored because of central nervous system *(CNS) toxicity and QT prolongation*, in addition to increases in hepatic enzymes. In addition, rilpivirine may cause depression and insomnia.

Protease Inhibitors

Protease inhibitors in general may cause diabetes or insulin resistance, hyperlipidemia, and dyslipidemia. Specifically, darunavir may cause severe rash and fever and elevated liver enzymes. Atazanavir has been known to be nephrotoxic. Boosting with ritonavir may limit these adverse effects.

Integrase Inhibitors

Integrase inhibitors do not usually cause adverse effects, but possibilities include worsening of psychiatric illness such as depression. They may also cause insomnia and

What are the potential adverse effects of integrase inhibitors?

What are the drug interactions associated with antiretrovirals? (LO#4)

In addition to specific drug side effects, some combinations of drugs may cause less than desirable interactions. An important concept to remember in antiretrovirals is CYP450 induction or inhibition. For example, efavirenz is an NNRTI that induces CYP450 and may decrease levels of certain drugs, such as bictegravir (integrase inhibitor cleared by CYP450).

CLINICAL CORRELATION

In Baltimore City in patients 13 years and older was 44.7 per 100,000 people, according to the city health department. Having such a high rate of new cases, Baltimore was among several cities targeted for federal strategies aimed at wiping out HIV by 2040. A study published in December 2018 found that delivering pre-exposure prophylaxis (PrEP) at sexually transmitted infection (STI) clinics to the vulnerable population of men who have sex with men in Baltimore is working to reduce the rate of new diagnoses. The study results, discussed in "Impact of providing pre-exposure prophylaxis for HIV at clinics for sexually transmitted infections in Baltimore City: an agent-based model," suggest that STI clinics could be the ideal place to broadly administer PrEP.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to MB, what do you say about whether he should take anti-HIV medications at this time?

You have treated many needlestick injuries; fortunately, they have become less common over the years. You contact the infectious disease specialist on call to confer about the level of risk in this individual case and whether to start a prophylactic regimen. You report back to MB, "The ID specialist and I both feel there is enough risk in this exposure that you should take anti-HIV medications as a preventative measure. When the source patient's HIV status is known, you and your physician can reassess the situation and decide whether to continue the medications." You answer MB's questions, he agrees to take postexposure prophylaxis, and you give MB his initial medication doses.

Summary

- Antiretroviral drug classes are based on specific steps in viral replication.
- Classes of antiretrovirals include entry inhibitors, nucleotide/nucleoside reverse transcriptase inhibitors, non-nucleoside reverse transcriptase inhibitors, protease inhibitors, and integrase inhibitors.
- NRTIs act as nucleotide/nucleoside analogues to inhibit reverse transcriptase, whereas NNRTIs inhibit the same enzyme by binding to it.
- Integrase strand transfer inhibitors block the integration of viral DNA into the host genome; protease inhibitors block the activity of protease, which cuts polyproteins into functional viral units.
- Drug combinations are used for antiretroviral therapy; generally, this consists of two NRTIs and one active antiretroviral from another class.
- Combining antiretrovirals that induce or inhibit CYP450 may change levels of other drugs used in combination.
- Pre-exposure and postexposure prophylaxis for HIV includes antiretrovirals; a three-drug regimen is used for postexposure prophylaxis.

Review Questions

1. Which drug class binds directly to reverse transcriptase?

A. ○ NRTIs
B. ● NNRTIs
C. ○ Protease Inhibitors
D. ○ Integrase

Hide Explanation

The correct answer is NNRTIs (B). NNRTIs bind directly to reverse transcriptase, inactivating it. NRTIs (A) inhibit reverse transcriptase by being incorporated into the growing nucleotide chain. Protease inhibitors (C) inhibit the protease enzyme, not reverse transcriptase. Integrase inhibitors (D) inhibit the integrase enzyme, not reverse transcriptase.

2. Which drug class is most commonly used with two NRTIs in initial HIV therapy?

A. ○ Protease inhibitors
B. ● INSTIs
C. ○ NNRTIs
D. ○ Fusion/ entry inhibitors

Hide Explanation

The correct answer is INSTI (B). Studies support the use of two NRTIs plus an INSTI for initial therapy. Protease inhibitors (A) and NNRTIs (C) are also options but less commonly used initially. Fusion/ entry inhibitors (D) are not typically first-line choices.

3. Which NRTI is rarely used due to pancreatitis risk?

A. ○ Tenofovir
B. ○ Zidovudine
C. ● Didanosine
D. ○ Efavirenz
E. ○ Rilpivirine

Hide Explanation

The correct answer is didanosine (C). Didanosine is rarely used because of the risk of pancreatitis. Tenofovir (A) may cause kidney injury or bone loss. Zidovudine (B) may cause bone marrow suppression. Efavirenz (D) is an NNRTI, not an NRTI. Rilpivirine (E) is also an NNRTI and may cause depression and insomnia.

4. Which drug is a CYP450 inducer?

A. ● Efavirenz
B. ○ Bictegravir
C. ○ Raltegravir
D. ○ Dolutegravir
E. ○ Atazanavir

Hide Explanation

The correct answer is efavirenz (A). Efavirenz is an NNRTI that induces CYP450 and may decrease levels of certain drugs. Bictegravir (B) is an integrase inhibitor cleared by CYP450, not an inducer. Raltegravir (C) and Dolutegravir (D) are also integrase inhibitors. Atazanavir (E) is a protease inhibitor, not a CYP450 inducer.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the mechanism of action and mechanism of resistance of antiretroviral agents used in the treatment and prevention of HIV infections.~~

02 ☑ ~~List the~~ ~~n the treatment and prevention HIV infections.~~

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03 ☑ ~~Recognize the common adverse reactions associated with drugs used in the treatment and prevention of HIV infections.~~

04 ☑ ~~Identify the common drug interactions associated with drugs used in the treatment and prevention of HIV infections.~~

Objective Confidence: 4/4